

Umang Singh

Backend software engineer, **2+ years** at **Oracle Financial Services**, building APIs, microservices and distributed systems in Java.

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WORK EXPERIENCE

Oracle (OFSS) – Machine Learning for Anti-Money Laundering Team (ML4AML)

Mumbai, India

Application Software Engineer

Jul 2024 – Present

- **Core developer** on the Automated Scenario Calibration (**ASC**) platform, Oracle's SaaS product for anti-money-laundering analytics, building Java microservices on Docker, Kubernetes, and OCI and **migrating its metering service** from Spring Boot to Helidon.
- Redesigned provisioning error handling across **6** Java integrations between ASC SaaS and its data-science platform, separating non-retryable failures from transient ones to eliminate up to **5 minutes** of fixed waiting per failure, while retaining **bounded exponential backoff**, status polling, idempotent duplicate handling, terminal-state propagation, and health checks.
- Implemented, SQL-registered and unit-tested **24 Python** behavioral red-flag feature transformers for AML behavioral models.
- Decomposed the **30,000+ line Python layer** (200+ APIs) through which data scientists' notebooks call the ML analytics, **separating database access from the analytics logic** so each part can be tested and changed on its own.
- Maintain the ML4AML services' **Liquibase** schema migrations, **Jenkins**/OCI CI/CD, and dependency remediation, plus the Compliance Studio notebooks used for calibration; built a bulk threshold-management workflow.

Project Intern

Jun 2023 – Jul 2023

- Constructed **25+ REST APIs** and containerized **Helidon** services with Docker for ASC SaaS, and established initial deployment pipelines, condensing **50+** paragraphs of analytical notebooks to streamline the model-calibration workflow for data scientists.

Futurepreneur Tech (StayWise)

Mumbai, India

Tech Lead

Feb 2023 – May 2023

- **Led a 15+ member** team across UI/UX, frontend, and backend; architected the Firebase backend, shipped the app in **90 days** to 20+ hostels, and secured **INR 100,000** in funding from the principal, the first startup at the college to do so.

EDUCATION

University of Mumbai | [TSEC Hall of Fame](#) (30 of 600+)

Mumbai, India

Bachelor of Engineering, Computer Engineering | **GPA: 8.76/10**

2020 – 2024

PROJECTS

[KrishiBodha](#): **Agentic AI Solution for Farmers** | GCP, Gemini, Gemma, Vertex AI, Google ADK, Flutter

Jun 2025 – Aug 2025

- Designed the agent orchestration layer in **Google ADK**, using Gemini to reason over a crop, market, and subsidy knowledge base and generate crop recommendations, with an on-device **Gemma** fallback that keeps the assistant answering when connectivity drops.
- Served the models on **Vertex AI** and unified multilingual voice input, image-based crop-disease diagnosis, localized market insights and alerts, and satellite crop-and-soil-health monitoring behind a single **Flutter** client usable offline.

[VidyuRakshak](#): **Satellite-Imagery-Based Transmission-Line Safety** | Google Earth Engine, ML, Flutter, AWS

Dec 2023 – Aug 2024

- Developed end-to-end geospatial ML pipelines for power-corridor risk analysis using Landsat and stereo imagery, training a **Random Forest Regressor** across **7 vegetation indices** and validating canopy heights against a **107,000-tuple** DTM/DSM dataset.
- [Selected by Power Grid Corporation of India for continued development under a signed agreement](#), funding a second proof of concept on AWS (S3, Lambda, API Gateway) for automated tree-growth monitoring and predictive analytics.

PUBLICATIONS

- [“Advanced Deep Learning Techniques for Early Detection of Oral Cancer: A Multimodal Approach”](#); ISAI 2025, Springer.

Mar 2025

- [“Exploring Challenges and Results in Crop Type Mapping Using Satellite Imagery”](#); IEEE InGARSS 2024. **First author.**

Dec 2024

- [“Detecting Tree Height and Addressing Challenges Along Power Line Corridors Using Optical and Stereo Satellite Imagery”](#); ICICBDA 2024, Springer.

Jun 2024

AWARDS AND ACHIEVEMENTS

- **Led a team to 1st Runner-Up** at [Google Agentic AI Hackathon 2025](#), a **Guinness World Record** event with **57,000+** registrants.
- **1st Runner-Up** at [Japan-India Hackathon 2024](#), organized by IIT Gandhinagar and NEDO Japan, with a prize of **INR 150,000**.
- **Winner** of [Smart India Hackathon 2023](#), a Government of India initiative and India's largest hackathon, with **250,000+** participants.
- **Top 8 of 11,000+** at the BIAL Genie Hackathon, powered by Kempegowda International Airport and Microsoft.
- Ranked **1,081 of 24,578 (top 4.4%)** in LeetCode [Weekly Contest 352](#), with **690+** problems solved across **500+** active days.
- **Mentored ~15 AI/ML teams** at [HackSync 2.0](#) (Google Developer Group TSEC); taught mathematics for two years at U&I NGO.

SKILLS

Languages & Backend: Java, Python, SQL, Kotlin, Helidon MicroProfile, Spring Boot, REST APIs, microservices, Flutter

Cloud & DevOps: OCI, GCP, AWS (Lambda, S3, API Gateway), Firebase, Docker, Kubernetes, Jenkins, Liquibase, Gradle, CI/CD, Git

AI/ML: LLMs, RAG, agentic workflows (Google ADK), Gemini, Gemma, Vertex AI, TensorFlow, Hugging Face

AI-assisted development: Claude Code, Codex, GitHub Copilot; prompt engineering, review and validation of generated code